

Chapter 9: Propagation of Light

Name: _____ Date: _____

Questions

1. According to the principle of rectilinear propagation, how does light travel in a homogeneous transparent medium?

- a) In zigzag patterns
- b) In straight lines
- c) In circular waves
- d) It bends randomly

Hint: Think about why you can't see around a corner without a mirror.

2. What do we call a single line representing the direction of light travel?

- a) A light beam
- b) A light ray
- c) A light wave
- d) A photon packet

Hint: A beam is a collection (bundle) of these.

3. You stand in front of a plane mirror. Which statement best describes the image formed?

- a) Real and larger than you
- b) Real and smaller than you
- c) Virtual and smaller than you
- d) Virtual and of the same size

Hint: Does your reflection look smaller or the same size? Can you project it onto a screen?

4. When an optical system (like a camera lens) forms an image on a sensor or film, that image is generally:

- a) Real and inverted
- b) Virtual and upright
- c) Virtual and magnified
- d) None of the above

Hint: To be captured on a physical surface (sensor), the light rays must actually meet there.

5. Light travels in a straight line unless it interacts with an object. Which of the following is NOT one of the three main interactions mentioned?

- a) Reflection
- b) Refraction
- c) Acceleration
- d) Absorption

Hint: Light speed is constant in a vacuum; it doesn't speed up.

6. Which of the following is a Luminous Object (Primary Source)?

- a) The Moon
- b) A Plane Mirror
- c) The Sun
- d) A White Piece of Paper

Hint: A luminous object produces its own light. The others just reflect it.

7. We are able to see non-luminous objects (like this worksheet) because:

- a) They emit their own light
- b) They refract light passing through them
- c) They reflect light towards our eyes
- d) They are transparent

Hint: If you turn off the lights in a room, these objects become invisible.

8. A light beam consisting of rays that are parallel to each other (like a laser) is called a:

- a) Converging beam
- b) Diverging beam
- c) Cylindrical (Parallel) beam
- d) Scattered beam

Hint: The width of this beam remains constant as it travels.

9. What do we call a beam of light where all the rays come together to meet at a single point?

- a) Converging beam
- b) Diverging beam
- c) Cylindrical beam
- d) Parallel beam

Hint: This happens when light passes through a magnifying glass.

10. Light rays spreading out from a small source (like a candle flame or car headlight) form

a:

- a) Converging beam
- b) Diverging beam
- c) Cylindrical beam
- d) Focused beam

Hint: The rays get farther apart as they travel away from the source.